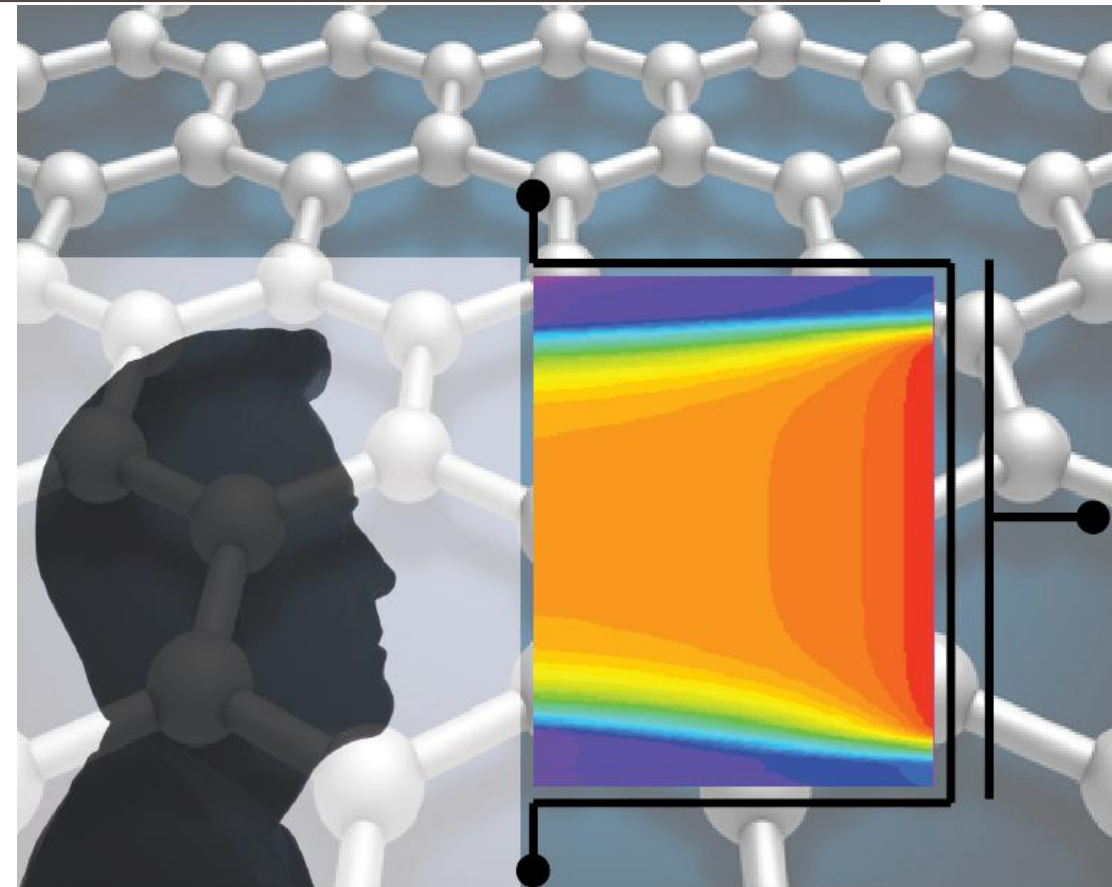


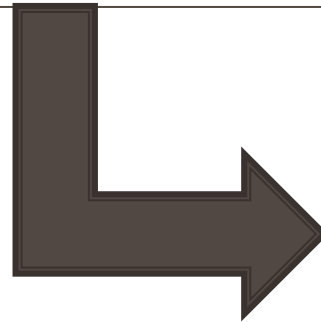
ADVANCED SOLID-STATE DEVICES

Week-14, Lecture-1

Sneh Saurabh
21st April, 2025

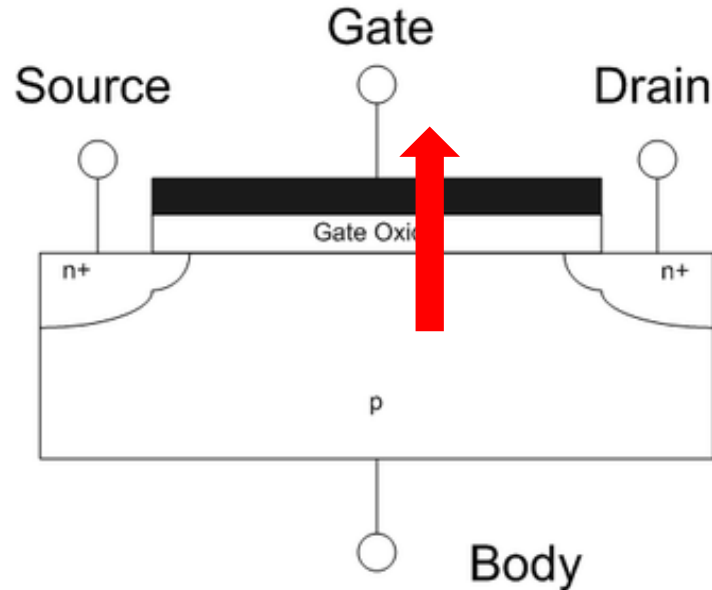


Advanced MOSFETs



High-k Dielectric

High-k Dielectric: Need at Advanced Technology Nodes



Effects of high *tunnel current* through *Gate Oxide*:

- Higher OFF-state power dissipation
- Reduced device reliability

Scaling MOSFETs and *Gate Oxide*:

- With scaling, the thickness of *Gate Oxide* is subsequently reduced

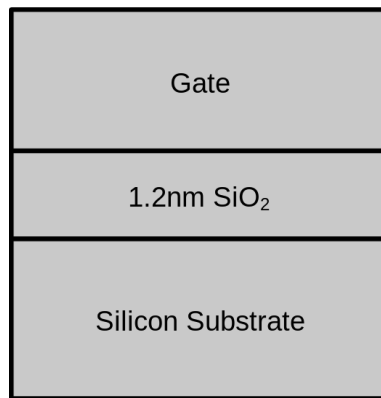
Effects of ultrathin *Gate Oxide*:

- The ultrathin *Gate Oxide* layer reduces the width of the tunneling potential barrier that separates the gate from the channel
- Reduced tunneling barrier width $\Rightarrow$ electrons/holes can easily tunnel through the *Gate Oxide*
- When *Gate Oxide* thickness is below 2 nm, *tunnel current through Gate Oxide* increases drastically

High-k Dielectric: To the rescue of Moore's Law

- High-k Dielectric: Materials with a high dielectric constant κ [kappa] as compared to *silicon dioxide*
 - Dielectric constant κ of a material measures its ability to hold charge
 - In a parallel plate capacitor, if air in the capacitor is replaced by a material with dielectric constant κ , then the capacitance will increase by κ times
-
- High-k dielectric materials have replaced silicon dioxide for *Gate Oxide* in MOSFETs in advanced CMOS technologies [45 nm and below]
 - Commonly used high-k dielectric materials are *hafnium dioxide* (HfO_2), *zirconium dioxide* (ZrO_2) and *titanium dioxide* (TiO_2)

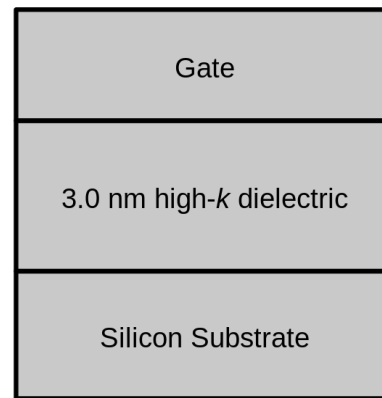
High-k Dielectric: Illustration



Existing 90nm Process
Capacitance = 1x
Leakage Current = 1x

$$\kappa = 3.9$$

Source: Wikipedia



A potential high-k process
Capacitance = 1.6x
Leakage Current = 0.01x

$$\kappa = 15.6$$

MOS capacitor can be modeled as: $C = \frac{\kappa\epsilon_0 A}{t_{ox}}$

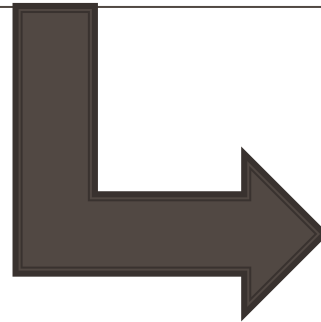
where $A \equiv \text{Capacitor Area}$

$\kappa\epsilon_0 \equiv \text{permittivity of gate oxide}$

$t_{ox} \equiv \text{thickness of gate oxide}$

- Drain Current: $I_D = \frac{\kappa\epsilon_0}{t_{ox}} \mu \left(\frac{W}{L} \right) \frac{(V_{GS} - V_T)^2}{2}$
- To increase drain current $\frac{\kappa\epsilon_0}{t_{ox}}$ has to be increased
- High-k dielectric helps in two ways:
 - Increase drain current
⇒ Speed/performance of circuit improves
 - Allows using thicker gate oxide [⇒ Lower Gate leakage Current]

Advanced MOSFETs



Strain
Engineering

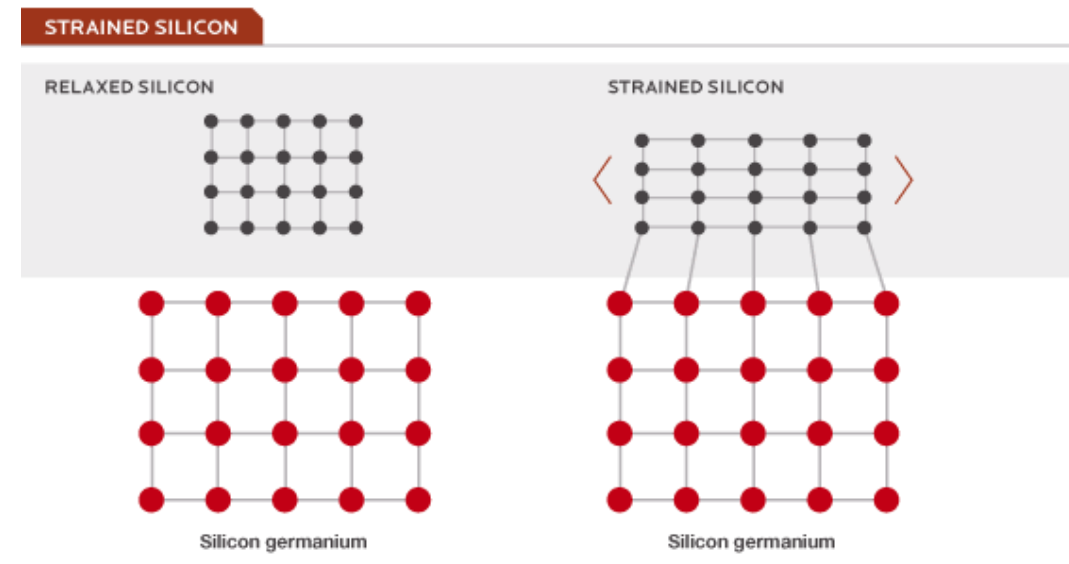
Strain Engineering: Basics

- Strain Engineering is a technique to improve the performance [speed] of an integrated circuit at nanoscale
- Strain Engineering has been used by AMD, IBM, Intel etc. in advanced process technologies [Intel has been using Strained Silicon in 90 nm technology and below]

- Strain: deforms the material
- Tensile Strain or Compressive Strain

Strain can be introduced by:

- Applying mechanical force: bending of wafer/substrate
- Lattice mismatch



Source: <http://www.asm.com/solutions/products/epitaxy-products/intrepid-xp-epitaxy>

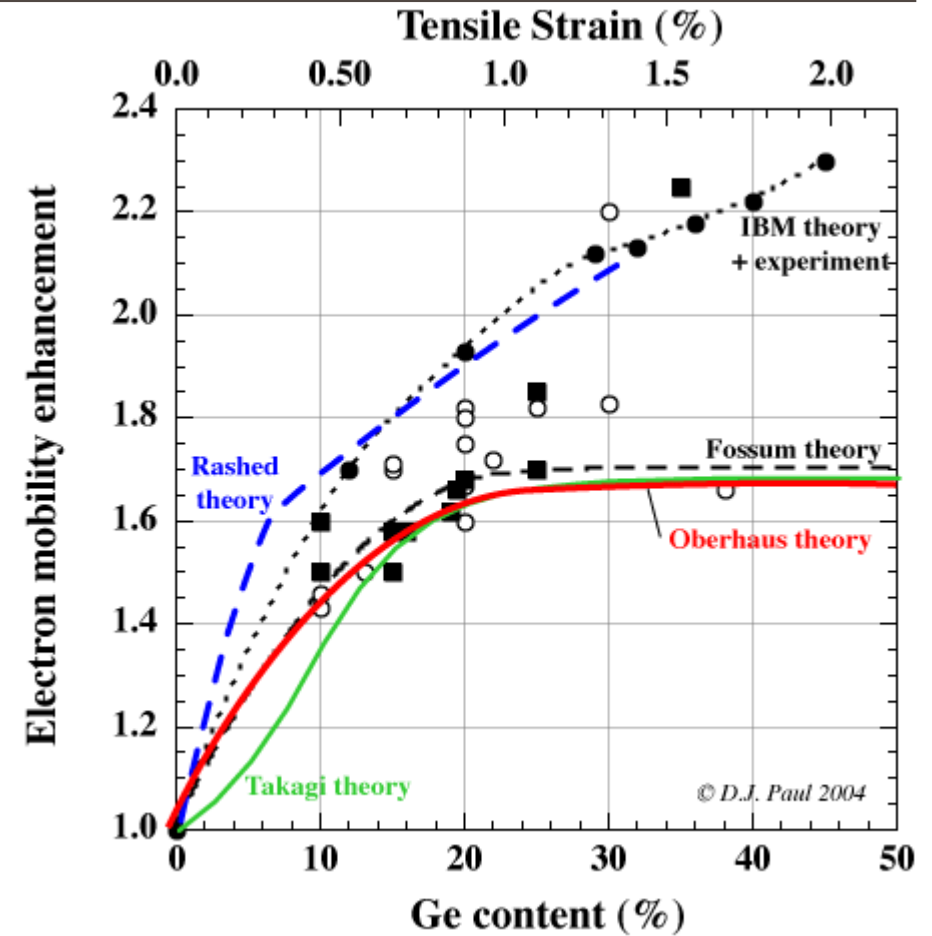
Strain Engineering: Impact

- Strain causes changes in lattice spacing of atoms
- Leads to increase in mobility of carriers [less hindrance in transport]

- Increase in mobility $\Rightarrow$ Drain Current increases

$$\triangleright I_D = \frac{\kappa \epsilon_0}{t_{ox}} \mu \left(\frac{W}{L} \right) \frac{(V_{GS} - V_T)^2}{2}$$

- Increase in Drain current $\Rightarrow$ Speed/Performance of devices/circuit improve



Source:
<http://userweb.eng.gla.ac.uk/douglas.paul/SiGe/electronics.html>

Steep Subthreshold Slope Devices

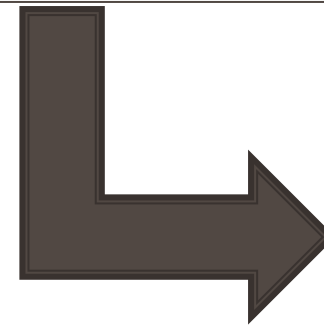
Steep Subthreshold Slope Devices: Future CMOS

- Conventional MOSFETs have subthreshold swing more than 60 mV/decade at 300K
 - Puts restriction on supply voltage

- Devices that work on some other *mechanism* and exhibit subthreshold swing *smaller than 60 mV/decade* are being actively explored

- Important Steep Subthreshold Slope Devices being explored:
 - Tunnel Field Effect Transistors [TFETs]
 - Negative Capacitance FETs [NCFET]
 - Impact Ionization MOSFET [IMOS]
 - Nanoelectromechanical Switches [NEMS]

Steep Subthreshold Slope Devices

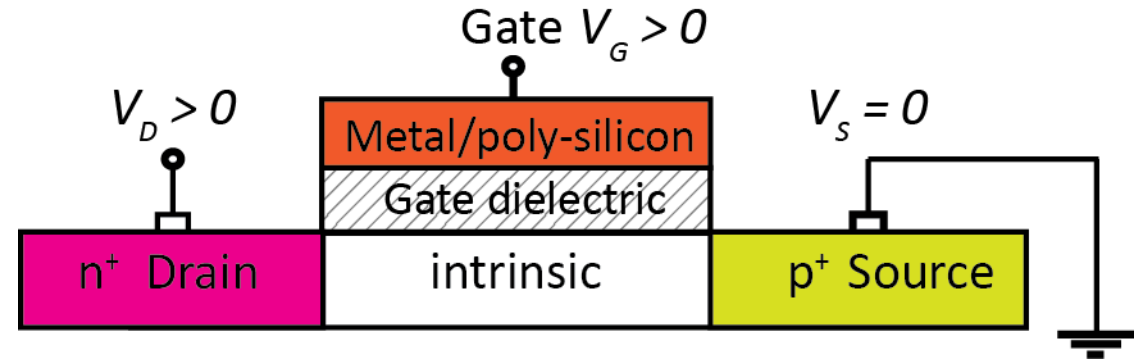


TFETs

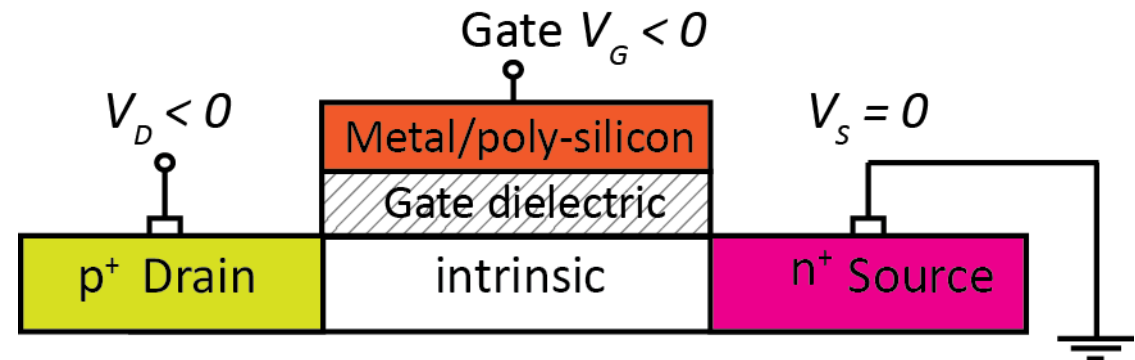
TFETs: Device Structure

- TFETs: Tunnel Field Effect Transistors
- Works on the basis of band-to-band tunneling

- In general, device structure is similar to a conventional MOSFET
- Doping of the source and the drain are of *opposite types* [in contrast to conventional MOSFET]
- Can be built using silicon, nanowire, CNT, Graphene etc.

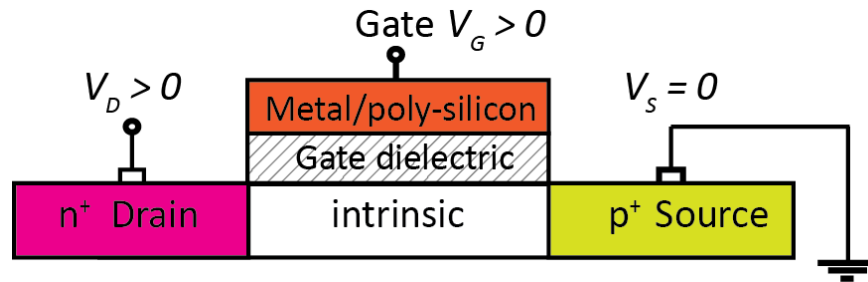


n-type TFET

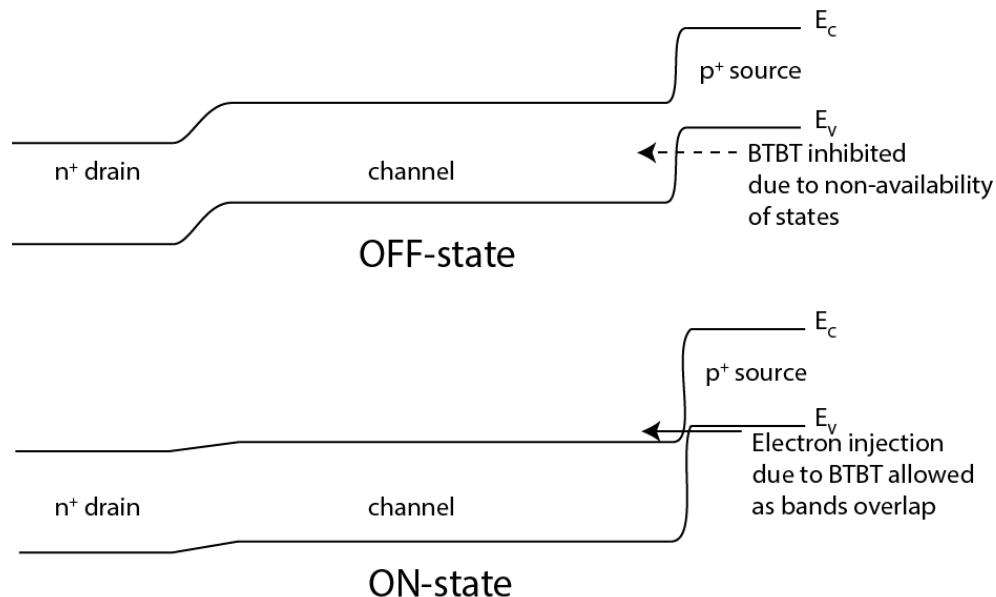


p-type TFET

TFETs: Operating principle

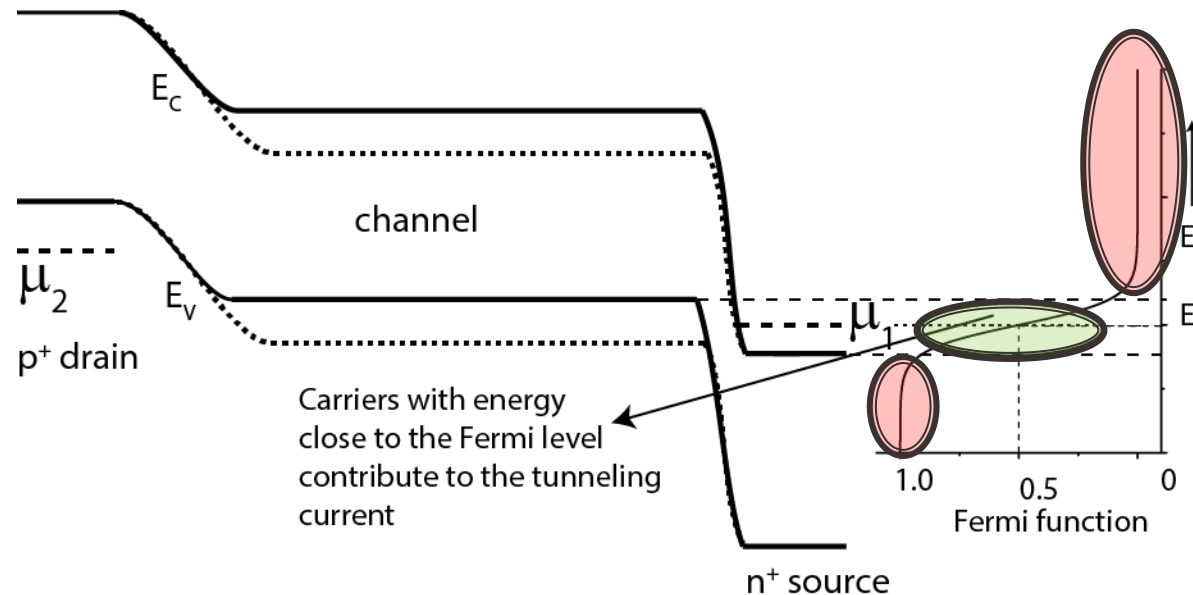


- Biasing of drain and source is done such that the transistor is similar to a reverse biased diode
- Normally very small current will flow in the OFF-state



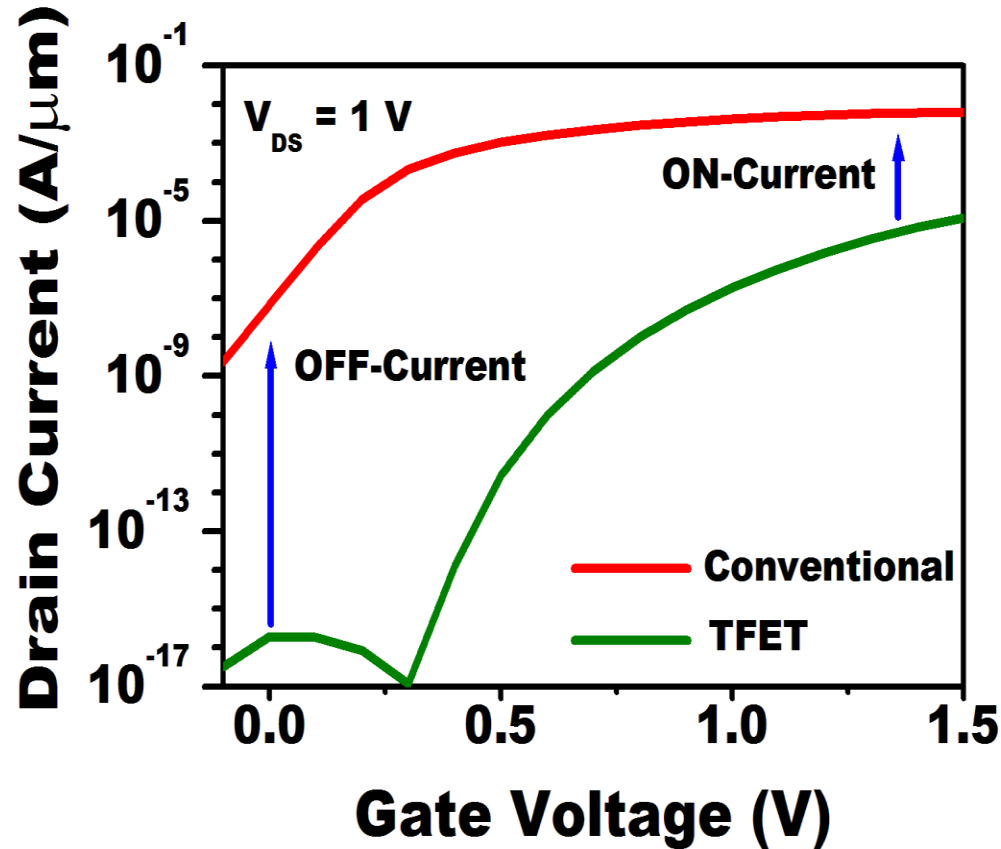
- Due to application of +ve gate bias, the energy levels in the channel are pushed downwards
 - Valence band in the source aligns with the conduction band in the channel
 - Electrons can tunnel from valence band in the source to conduction band in the channel
 - These electrons are pulled towards drain by +ve drain bias

TFETs: Subthreshold Swing



- Electrons in high energy tail do not contribute to the current
 - No states are available in the channel due to band gap
 - Subthreshold swing not limited to 60 mV/decade at room temperature
- Only electrons in energy range close to the Fermi Energy contribute to the current
 - The change in occupancy is sharp around Fermi Energy
 - This gives rise to subthreshold swing less than 60 mV/decade at room temperature

TFETs: Merits, Demerits and Challenges



Merits

- Subthreshold swing lower [subthreshold swing increases with gate voltage in contrast to conventional MOSFET]
- Very low OFF-state current

Demerits

- Low ON-state current

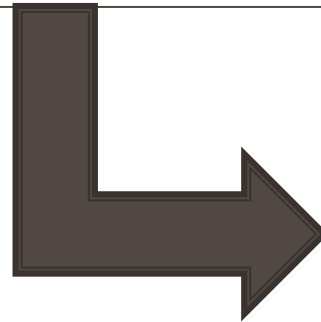
Challenges

- To obtain good ON-state current
- Obtain average subthreshold swing better than $60 mV/decade$ for more than 4 decades of current

TFETs: Recent Developments

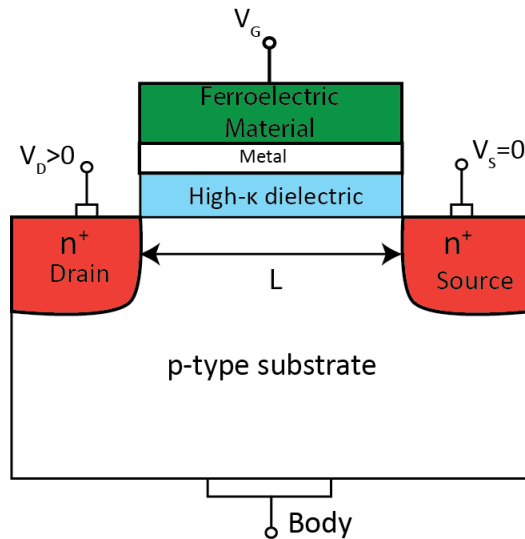
- ON-state currents have shown improvements using III-V materials
- Average subthreshold swing demonstrated to be better than 60 mV/decade for more than 4 decades of current using Transition Metal Dichalcogenides (TMD) materials
- Useful for energy-efficient low frequency applications (such as neuromorphic computing)

Steep Subthreshold Slope Devices



NCFETs

NCFETs: Device Structure



n-type NCFET



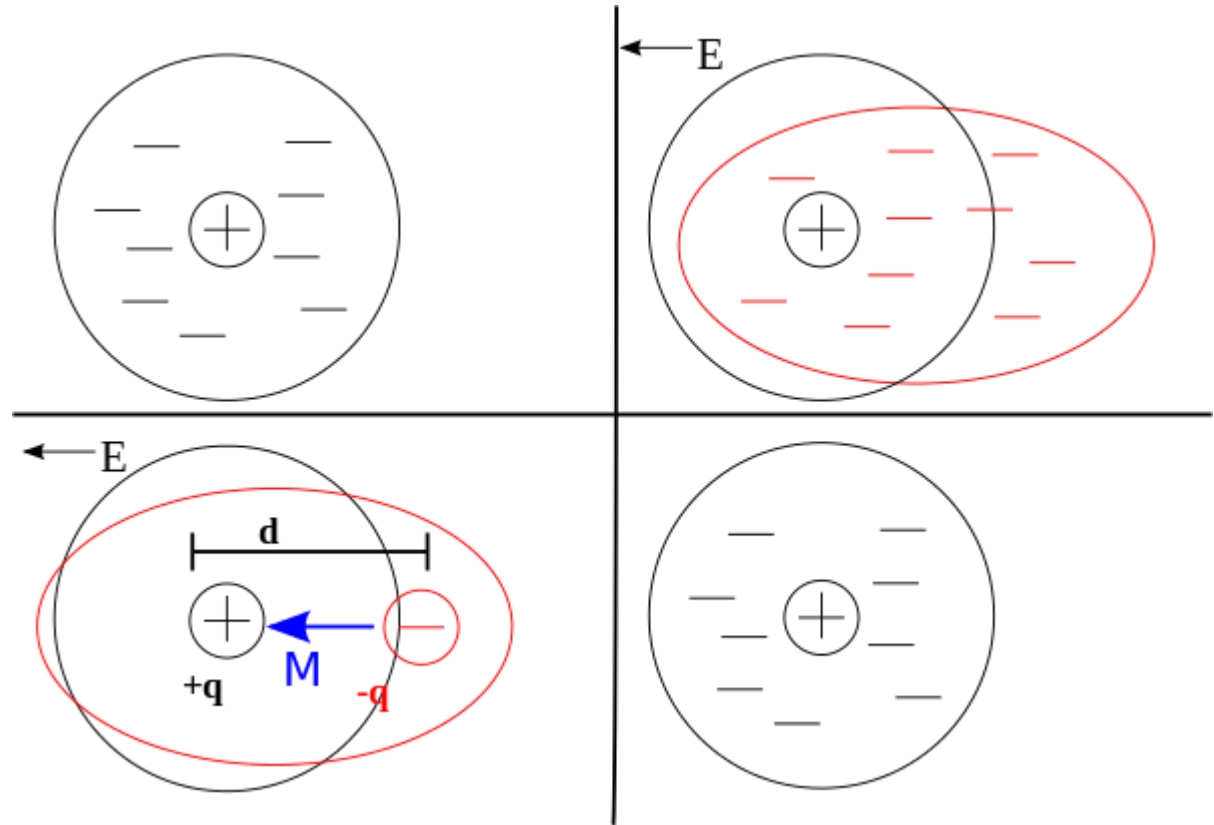
- NCFETs: Negative Capacitance Field Effect Transistors

- Device structure similar to a conventional MOSFET
- Conventional insulator in the gate of a MOSFET replaced with a combination of a *ferroelectric insulator* and a *high-k dielectric*

- Example:
 - PZT [$\text{Pb}(\text{Zr}_{0.2}\text{Ti}_{0.8})\text{O}_3$] used as ferroelectric insulator and STO [SrTiO_3] used as high-k dielectric
 - LaAlO_3 used as ferroelectric insulator and STO [SrTiO_3] used as high-k dielectric
- Thicknesses of ferroelectric insulator and high-k dielectric materials should be chosen appropriately [to obtain negative capacitance]

NCFETs: Polarization

- When a *dielectric material* is placed in an electric field electric dipole moments are induced
- Molecules gain electric dipole moment due to relative shift of the positive and negative charge
- It is measured as dipole moment induced per unit volume (denoted as $\vec{P} = Nq\vec{d}$)



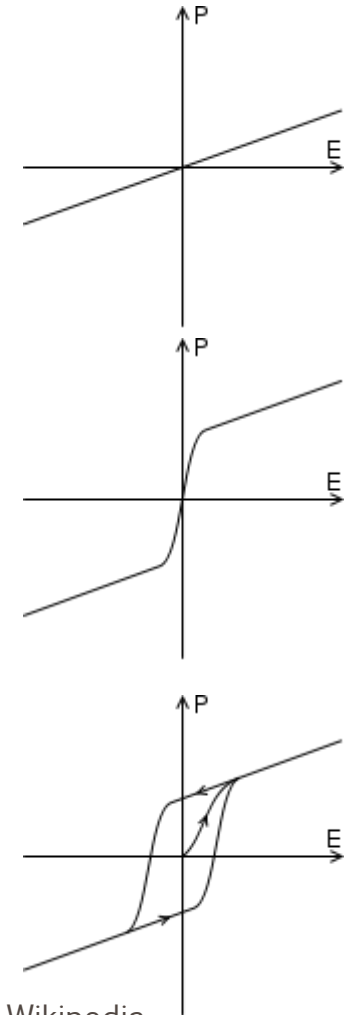
Source: Wikipedia

NCFETs: Ferroelectric Material

Dielectric Material: $\vec{P}$ is linearly related to electric field $\vec{E}$

Paraelectric Material: $\vec{P}$ is non-linearly related to electric field $\vec{E}$

Ferroelectric Material: $\vec{P}$ is non-linearly related to electric field $\vec{E}$ and a non-zero $\vec{P}$ remains even when $\vec{E} = \mathbf{0}$ [hysteretic behavior]



Source: Wikipedia

NCFETs: Origin of Negative Capacitance

- The capacitance can be expressed as [using minimization of total energy]:

$$C = \frac{1}{\left(\frac{d^2U}{dQ^2}\right)}$$

where $U \equiv \text{Total Energy}$, $Q \equiv \text{Charge on the Capacitor}$

- For non-linear capacitor energy can be given as:

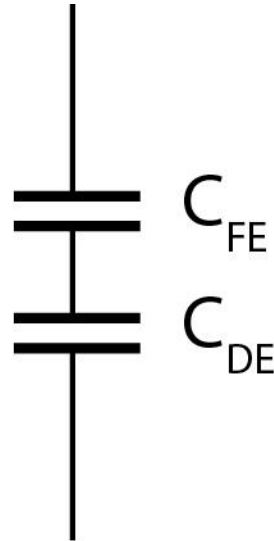
$$U = \alpha Q^2 + \beta Q^4 + \gamma Q^6$$

- For Ferroelectric capacitor: α is negative for temperature less than **Curie Temperature** [T_C]:

$$\alpha = \alpha_0(T - T_C)$$

- $\Rightarrow$ Ferroelectric capacitor can be biased in negative capacitance regime for certain temperatures and Q

NCFETs: Increased Total Capacitance



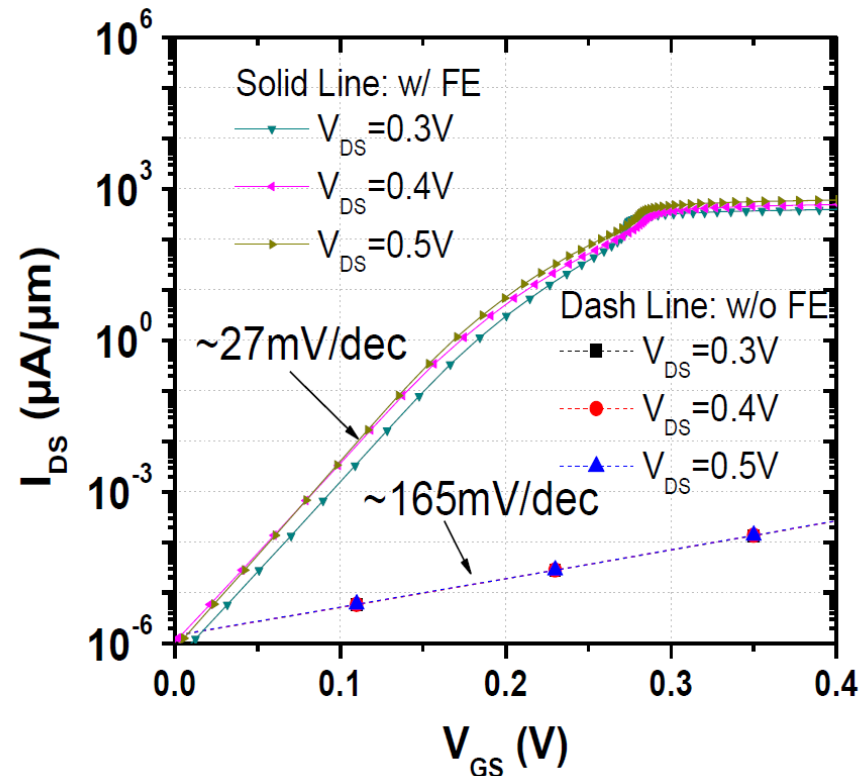
- In a series combination of a negative and a positive capacitor, the total capacitance becomes larger than its constituent positive capacitor
- This is just the opposite of what happens in a classical series combination of two positive capacitors, where the total capacitance is always reduced

NCFETs: Consequences of Negative Capacitance

- For a MOSFET, a negative gate capacitance can make the total capacitance, looking into the gate, larger than the semiconductor capacitance
 - $\Rightarrow$ To induce the same amount of charge in the channel smaller gate voltage is required
 - $\Rightarrow$ Smaller subthreshold swing can be obtained

- This is kind of *step-up transformer action* [amplification of applied gate voltage]

NCFETs: Characteristics, Merits/Demerits



Source: Yeung, Chun Wing, et al. "Non-Hysteretic Negative Capacitance FET with Sub-30mV/dec Swing over 106X Current Range and I_{ON} of 0.3 mA/ μm without Strain Enhancement at 0.3 V VDD." *Conf. Simul. Semicond. Processes Devices*. 2012.

Merits

- Direct replacement for conventional MOSFET since only the gate stack needs to be modified
- Operating principle and scalability similar to a MOSFET

Demerits/Challenges

- Identification of appropriate ferroelectric material and integration of that material into the conventional CMOS technology
 - Stability of ferroelectric material
 - Non-hysteretic operation

References

- S.Saurabh and M. J. Kumar, [Fundamentals of Tunnel Field Effect Transistors](#), *CRC Press (Taylor & Francis)*, ISBN 9781498767132